Light

GCSE Computer Science - Character Sets (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is a character set?

- ○ **A.** A collection of characters that a computer can recognise and use
- ○ **B.** A type of keyboard used for typing
- ○ **C.** A program that displays text on screen
- ○ **D.** A method of compressing text files

Q2 (1 mark)

What does ASCII stand for?

- ○ **A.** Automatic System Code for Internet Integration
- ○ **B.** Advanced Symbol Coding for Input Interpretation
- ○ **C.** Alphabetic Sequence Code for Internal Indexing
- ○ **D.** American Standard Code for Information Interchange

Q3 (1 mark)

How many bits are used to represent each character in 7-bit ASCII?

- ○ **A.** 7 bits
- ○ **B.** 4 bits
- ○ **C.** 8 bits
- ○ **D.** 16 bits

Q4 (1 mark)

How many different characters can be represented using 7-bit ASCII?

- ○ **A.** 64
- ○ **B.** 256
- ○ **C.** 512
- ○ **D.** 128

Q5 (1 mark)

In ASCII, the character 'A' has the decimal code 65. What type of characters are represented by codes 65 to 90?

- ○ **A.** Lowercase letters a to z
- ○ **B.** The digits 0 to 9
- ○ **C.** Uppercase letters A to Z
- ○ **D.** Punctuation marks

Q6 (1 mark)

Why are ASCII character codes arranged in sequential groups?

- ○ **A.** To allow programmers to calculate codes for other characters in the same group
- ○ **B.** To make the codes look more attractive when displayed
- ○ **C.** To reduce the file size of text documents
- ○ **D.** To make typing faster on a keyboard

Q7 (1 mark)

What is the main purpose of Unicode?

- ○ **A.** To speed up internet connections
- ○ **B.** To compress text files to save storage space
- ○ **C.** To encrypt text for security purposes

- ○ **D.** To represent characters from many different languages and writing systems

Q8 (1 mark)

What is the relationship between Unicode and ASCII for character codes 0 to 127?

- ○ **A.** Unicode uses the same codes as ASCII for characters 0 to 127
- ○ **B.** Unicode uses completely different codes from ASCII
- ○ **C.** ASCII codes are exactly double the Unicode codes
- ○ **D.** Unicode only includes codes from 128 onwards

Q9 (1 mark)

In ASCII, lowercase 'a' has the code 97. What is the code for lowercase 'b'?

- ○ **A.** 66
- ○ **B.** 98
- ○ **C.** 96
- ○ **D.** 107

Q10 (1 mark)

Which of the following is an advantage of Unicode over ASCII?

- ○ **A.** Unicode uses fewer bits per character
- ○ **B.** Unicode can only represent English characters
- ○ **C.** Unicode files are always smaller than ASCII files
- ○ **D.** Unicode can represent a much larger number of different characters

Q11 (1 mark)

How does a computer store the letter 'A' internally?

- ○ **A.** As the binary code for its character set value
- ○ **B.** As a tiny picture of the letter
- ○ **C.** As the sound of someone saying the letter
- ○ **D.** As a written description of the letter

Q12 (1 mark)

In ASCII, the digit '0' has the code 48. What is the code for the digit '5'?

- ○ **A.** 5
- ○ **B.** 50
- ○ **C.** 55
- ○ **D.** 53

Q13 (2 marks)

In ASCII, the uppercase letter 'A' has the decimal code 65. What is the decimal code for the uppercase letter 'G'?

- ○ **A.** 70
- ○ **B.** 71
- ○ **C.** 72
- ○ **D.** 103

Q14 (2 marks)

In ASCII, lowercase 'a' has the code 97. What is the code for lowercase 'f'?

- ○ **A.** 102
- ○ **B.** 70
- ○ **C.** 101
- ○ **D.** 112

Q15 (2 marks)

A student looks up an ASCII table and finds that code 72 represents a character. Given that 'A' = 65, which character does code 72 represent?

- ○ **A.** F
- ○ **B.** G
- ○ **C.** H

- D. |

Q16 (2 marks)

A software company is developing an application for users in China, Japan, and the UK. Why would ASCII be unsuitable for this application?

- A. ASCII is too slow for modern computers
- B. ASCII files are too large for international use
- C. ASCII only works on American computers
- D. ASCII cannot represent Chinese and Japanese characters

Q17 (2 marks)

In ASCII, 'A' = 65 and 'a' = 97. What is the difference between the uppercase and lowercase codes for any letter?

- A. 26
- B. 32
- C. 52
- D. The difference varies for each letter

Q18 (2 marks)

A text file contains the word 'CAT' stored using 7-bit ASCII. How many bits are needed to store this word?

- A. 7 bits

- **B.** 21 bits
- **C.** 14 bits
- **D.** 24 bits

Q19 (2 marks)

A student wants to include mathematical symbols (such as Σ, ∞, and π) in their document. Why would Unicode be necessary rather than ASCII?

- **A.** ASCII files cannot be saved on modern computers
- **B.** Unicode processes mathematical calculations faster
- **C.** ASCII does not include these mathematical symbols in its character set
- **D.** Unicode automatically performs the calculations

Q20 (2 marks)

The ASCII code for 'D' is 68. What is this value in binary?

- **A.** 0100 0100
- **B.** 0100 0010
- **C.** 0100 0101
- **D.** 0100 1000

Q21 (2 marks)

A company has many old documents stored in ASCII format. They want to upgrade their systems to use Unicode. Why will their old documents still work correctly?

- ○ **A.** Unicode automatically deletes ASCII files
- ○ **B.** Unicode software cannot read any ASCII files
- ○ **C.** Unicode uses the same codes as ASCII for the first 128 characters
- ○ **D.** All old documents must be manually converted

Q22 (2 marks)

A programmer knows that in ASCII, 'M' = 77. Without using a table, what is the code for 'P'?

- ○ **A.** 79
- ○ **B.** 81
- ○ **C.** 87
- ○ **D.** 80

Q23 (2 marks)

ASCII includes 'control characters' with codes 0 to 31. What is the purpose of these characters?

- ○ **A.** They represent uppercase letters
- ○ **B.** They control text formatting and device operations, such as new lines and tabs
- ○ **C.** They represent mathematical operators

- ○ **D.** They are unused codes reserved for future expansion

Q24 (3 marks)

In ASCII, 'a' = 97. A programmer wants to determine if a character code represents a lowercase letter. Which condition correctly checks if a code (stored in variable `code`) represents a lowercase letter?

- ○ **A.** code >= 65 AND code <= 90
- ○ **B.** code >= 97 AND code <= 122
- ○ **C.** code >= 48 AND code <= 57
- ○ **D.** code >= 0 AND code <= 127

Q25 (3 marks)

Compare ASCII and Unicode by considering: (1) the number of characters each can represent, (2) the number of bits typically used, and (3) their suitability for international websites.

- ○ **A.** ASCII represents more characters, uses fewer bits, and is better for international websites
- ○ **B.** ASCII and Unicode are identical in all three aspects
- ○ **C.** Unicode represents fewer characters but is faster to process
- ○ **D.** ASCII represents 128 characters using 7 bits and is limited to English; Unicode represents over 140,000 characters, uses 8-32 bits, and supports virtually all world languages making it essential for international websites

Q26 (3 marks)

A programmer needs to convert a lowercase letter to uppercase. Given that 'A' = 65 and 'a' = 97, and knowing the difference is consistent for all letters, which operation correctly converts a lowercase letter code to its uppercase equivalent?

- ○ **A.** Subtract 32 from the lowercase code
- ○ **B.** Add 32 to the lowercase code
- ○ **C.** Multiply the lowercase code by 2
- ○ **D.** Divide the lowercase code by 2

Q27 (3 marks)

A document contains 1000 characters of English text. Calculate the storage difference between saving it as 7-bit ASCII versus 16-bit Unicode.

- ○ **A.** ASCII: 1000 bytes, Unicode: 1000 bytes - no difference
- ○ **B.** ASCII: 875 bytes, Unicode: 2000 bytes - Unicode needs 1125 bytes more
- ○ **C.** ASCII: 7000 bits, Unicode: 16000 bits - Unicode needs 9000 bits more
- ○ **D.** ASCII: 7000 bytes, Unicode: 16000 bytes - Unicode needs 9000 bytes more

Q28 (3 marks)

A string contains the characters 'CODE' stored in ASCII. Given that 'A' = 65, work out the sum of all four ASCII codes.

- A. 283
- B. 286
- C. 280
- D. 294

Q29 (3 marks)

Modern messaging applications allow users to send emoji (such as 😀 and ❤️) alongside regular text. Explain why these applications must use Unicode rather than ASCII.

- A. ASCII includes all emoji but processes them more slowly than Unicode
- B. Emoji require sound processing which only Unicode supports
- C. ASCII's 128 characters cannot include emoji, but Unicode's extensive character set includes thousands of emoji as standard characters with unique codes
- D. Emoji are stored as small image files and do not use character encoding at all

Q30 (3 marks)

A web developer is creating a website that will display content in English, Arabic, and Chinese. Analyse the character encoding requirements and explain why a single encoding standard is important.

- A. Three separate websites are needed, each using a different encoding standard for each language
- B. ASCII can handle all three languages if the correct font is installed

○ **C.** Unicode provides a single encoding standard that includes English, Arabic, and Chinese characters, ensuring consistent display across all browsers and devices, and allowing mixed-language content on the same page without encoding conflicts

○ **D.** The website only needs ASCII because browsers automatically translate characters

Total: **55** marks • Time: **30 min**

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